

CJRB-201, a promising candidate for microbiome-based therapy with potent anti-inflammatory effects in Inflammatory Bowel Disease

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Background

Inflammatory bowel disease (IBD) is a chronic inflammatory gastrointestinal condition characterized by immune dysregulation driven by microbiome dysbiosis, with a reduction of beneficial bacteria such as *Faecalibacterium*, a dominant commensal bacterium comprising 5-10% of the human gut microbiome. Its depletion is a hallmark of Crohn's disease and other autoimmune-related conditions, highlighting its potential as a new therapeutic modality.

Methods

CJRB-201 was selected using our BI platform (Ez-Mx) based on more than 500 metagenomic datasets from IBD patients and healthy controls. The immune regulatory efficacy of CJRB-201 was evaluated through *in vitro* and *ex vivo* assays, focusing on key cytokine ratios (IL-10/IL-1 β , IL-10/TNF, IL-10/IFN- γ) in macrophages and T cells. The induction of regulatory T cells (Tregs) was evaluated *ex vivo* and in germ-free (GF) mice. Three pathogenetically different IBD animal models (DSS-induced, IL-10 KO GF, T cell transfer) were examined to assess its efficacy. Lastly, the SHIME system (Simulator of the Human Intestinal Microbial Ecosystem) was applied to test various fibers that promote the growth of CJRB-201.

Results

Among 34 tested anti-inflammatory strains, including those from CJ Bioscience, 4D Pharma, and our Ez-Mx platform, CJRB-201 emerged as one of the most effective. Notably, CJRB-201 and its conforming species showed the highest induction of Tregs *ex vivo* among the sixty *Faecalibacterium* strains, primarily driven by its cell pellet form rather than the supernatant. Administration of CJRB-201 resulted in successful colonization in the colon of GF mice, leading to the enhanced Tregs. In the three IBD models with both preventative and therapeutic approaches, CJRB-201 demonstrated significant improvements across five key disease features (body weight, disease activity index, colon length, histopathology, and inflammatory markers such as cytokines and fecal lipocalin-2). Interestingly, CJRB-201 outperformed other potential single and consortia strains and demonstrated efficacy comparable to the α -IL-12p40 antibody, particularly in the T cell transfer model. Lastly, we identified the fiber that stabilizes the enrichment of CJRB-201 in the SHIME system.

Conclusion

CJRB-201 exhibits superior anti-inflammatory efficacy in multiple immune assays and IBD animal models, significantly improving preclinical and inflammatory markers. Given its ability to enhance Tregs and the promising role of the identified fiber in supporting its growth, CJRB-201 holds strong promise as a microbiome-based therapeutic drug for IBD patients.

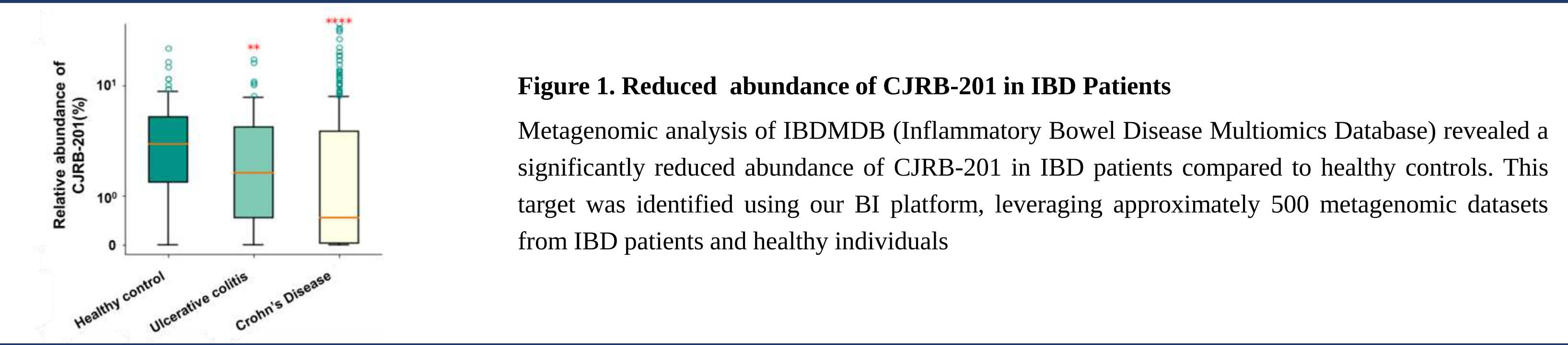


Figure 1. Reduced abundance of CJRB-201 in IBD Patients

Metagenomic analysis of IBDMDB (Inflammatory Bowel Disease Multiomics Database) revealed a significantly reduced abundance of CJRB-201 in IBD patients compared to healthy controls. This target was identified using our BI platform, leveraging approximately 500 metagenomic datasets from IBD patients and healthy individuals

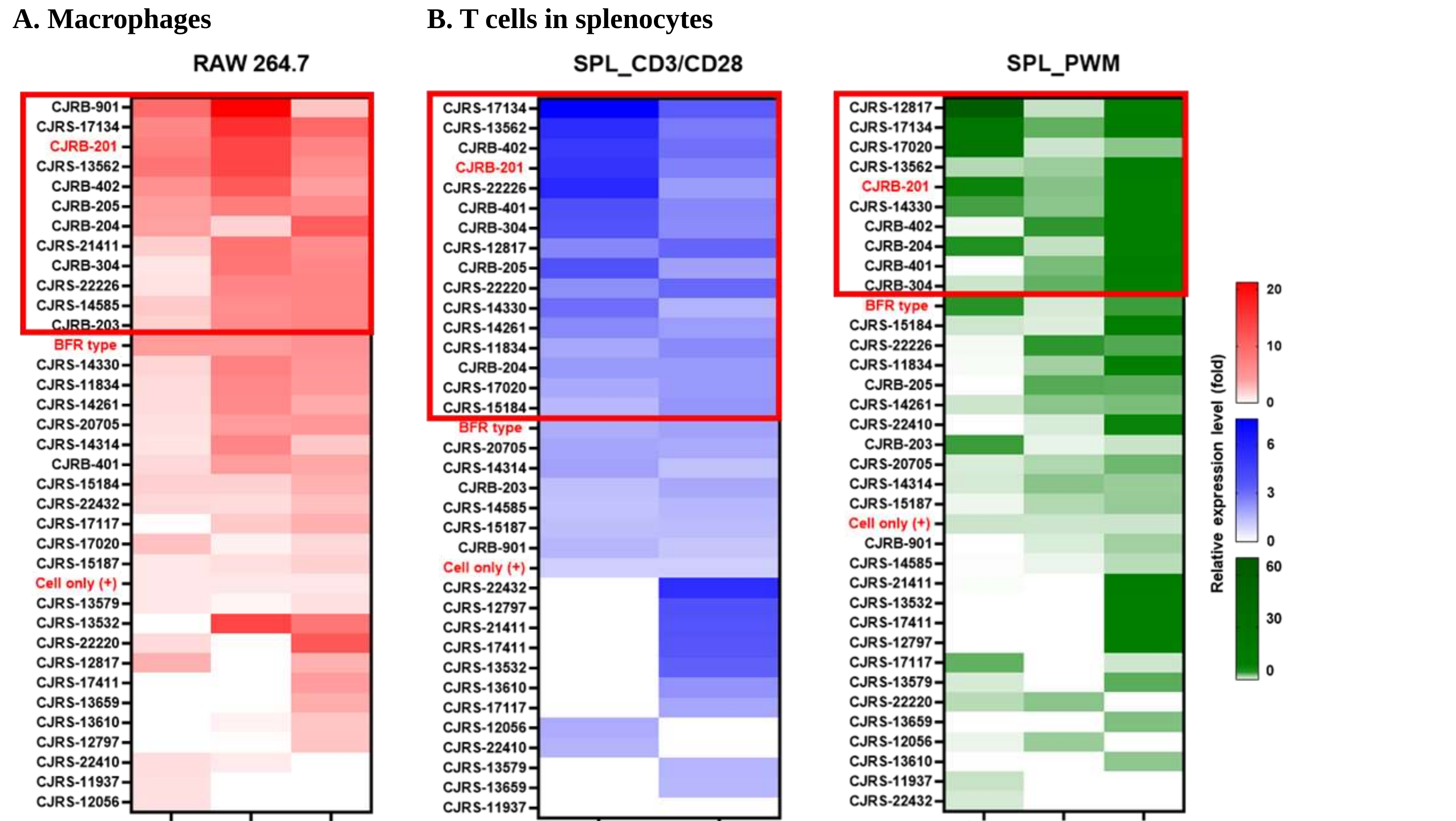


Figure 2. Comparative evaluation of strain efficacy *in vitro* and *ex vivo*. A total of 34 strains (6 from CJBS, 6 from 4D Pharma, and 22 from Ez-Mx) were tested via IL-10 based innate and adaptive immune screening, confirming that the CJRB-201 strain ranks among the top in terms of anti-inflammatory efficacy. **A.** After stimulating macrophages (RAW 264.7) with LPS, a comparative evaluation of IL-10/IL-1 β and IL-10/TNF was conducted. **B.** After stimulating T cells in splenocytes with CD3/CD28 or PWM, a comparative evaluation of IL-10 and IL-10/IFN- γ was conducted. Positive control: BFR type

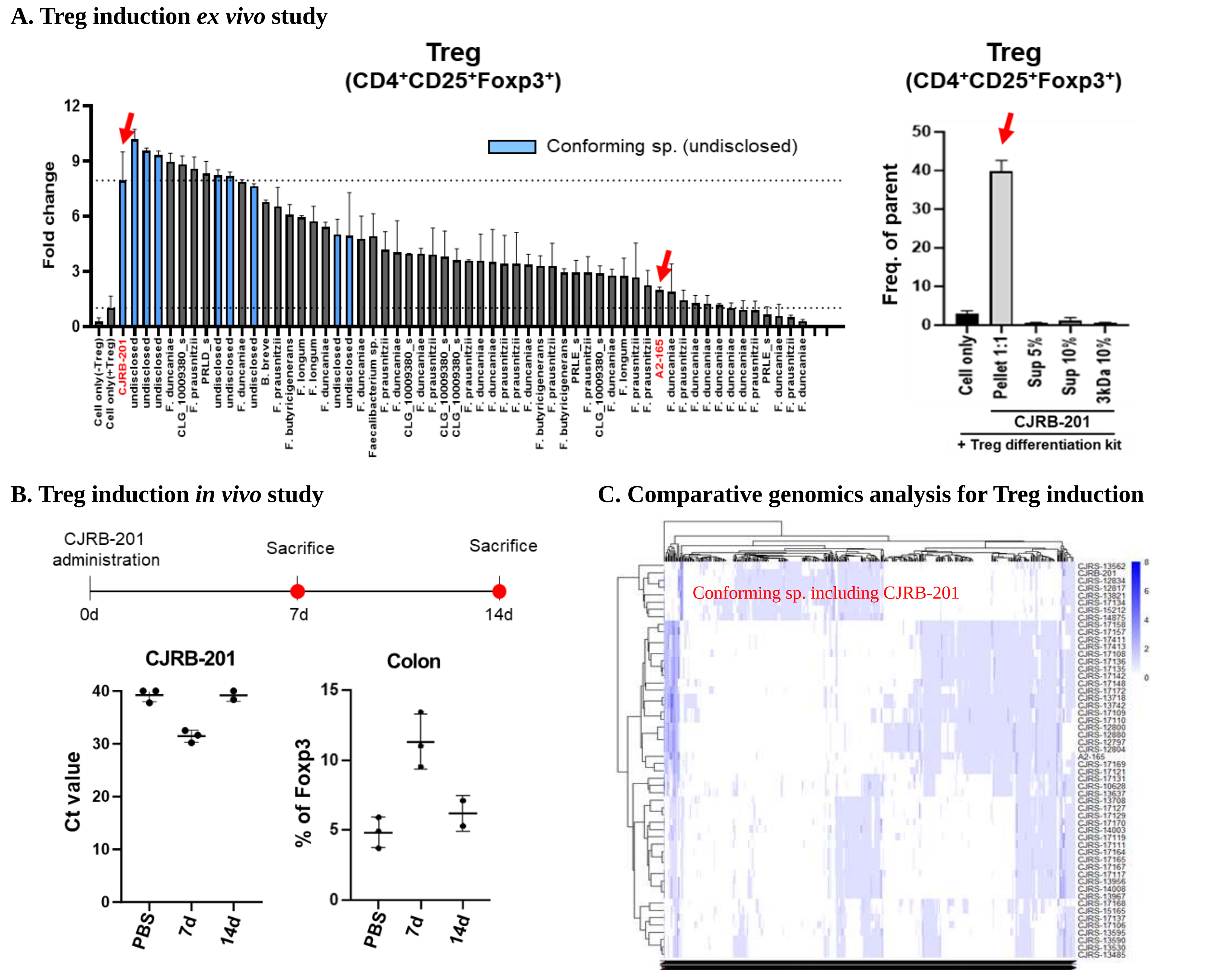


Figure 3. CJRB-201 promotes significant regulatory T cells (Tregs) *ex vivo* and *in vivo*. **A.** CJRB-201, along with its conforming species (undisclosed), demonstrates the highest increase in Tregs, primarily induced by cell pellets rather than the cultured supernatant. **B.** The experimental scheme of germ-free mice treated with a single dose of CJRB-201 on day 0, followed by weekly sacrifices for analysis. CJRB-201 administration successfully colonized the germ-free mouse colon and enhanced Tregs. **C.** Comparative genomics analysis of *Faecalibacterium* strains tested for Treg induction (A) reveals that the CJRB-201 species possesses unique genomic features essential for Treg induction.

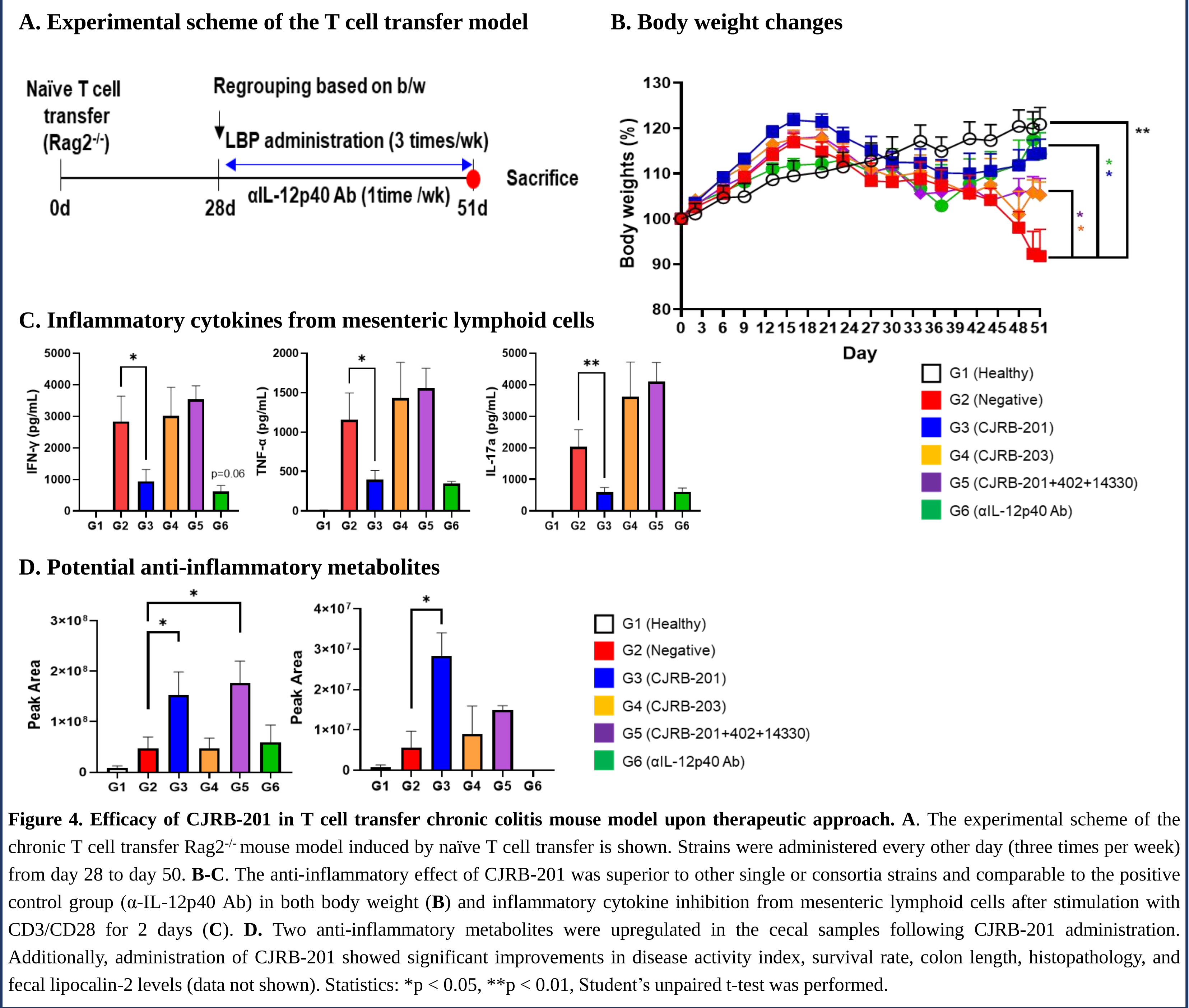


Figure 4. Efficacy of CJRB-201 in T cell transfer chronic colitis mouse model upon therapeutic approach. **A.** The experimental scheme of the chronic T cell transfer Rag2^{-/-} mouse model induced by naive T cell transfer is shown. Strains were administered every other day (three times per week) from day 28 to day 50. **B-C.** The anti-inflammatory effect of CJRB-201 was superior to other single or consortia strains and comparable to the positive control group (α -IL-12p40 Ab) in both body weight (**B**) and inflammatory cytokine inhibition from mesenteric lymphoid cells after stimulation with CD3/CD28 for 2 days (**C**). **D.** Two anti-inflammatory metabolites were upregulated in the cecal samples following CJRB-201 administration. Additionally, administration of CJRB-201 showed significant improvements in disease activity index, survival rate, colon length, histopathology, and fecal lipocalin-2 levels (data not shown). Statistics: *p < 0.05, **p < 0.01, Student's unpaired t-test was performed.

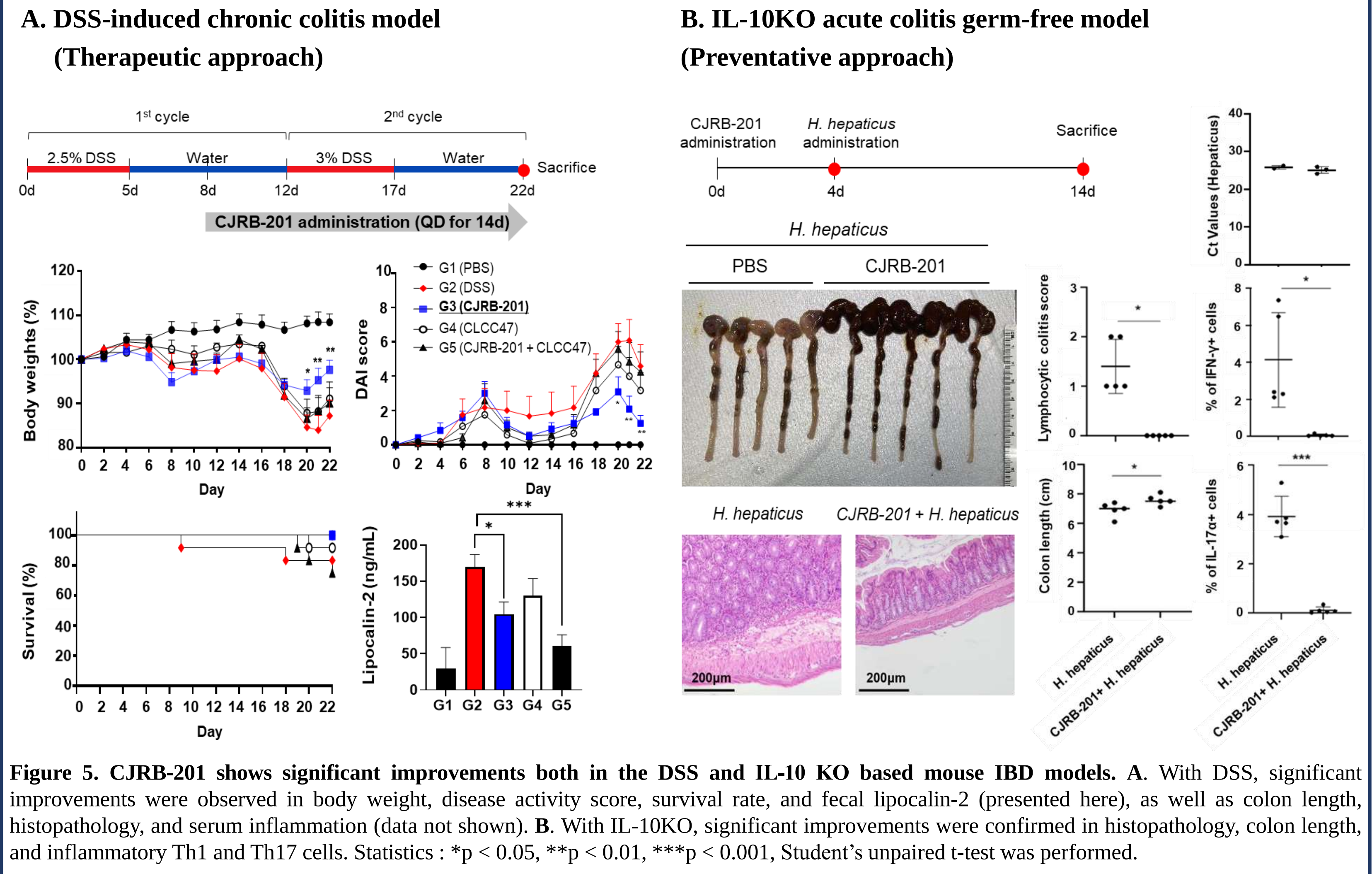


Figure 5. CJRB-201 shows significant improvements both in the DSS and IL-10 KO based mouse IBD models. **A.** With DSS, significant improvements were observed in body weight, disease activity score, survival rate, and fecal lipocalin-2 (presented here), as well as colon length, histopathology, and serum inflammation (data not shown). **B.** With IL-10KO, significant improvements were confirmed in histopathology, colon length, and inflammatory Th1 and Th17 cells. Statistics: *p < 0.05, **p < 0.01, ***p < 0.001, Student's unpaired t-test was performed.

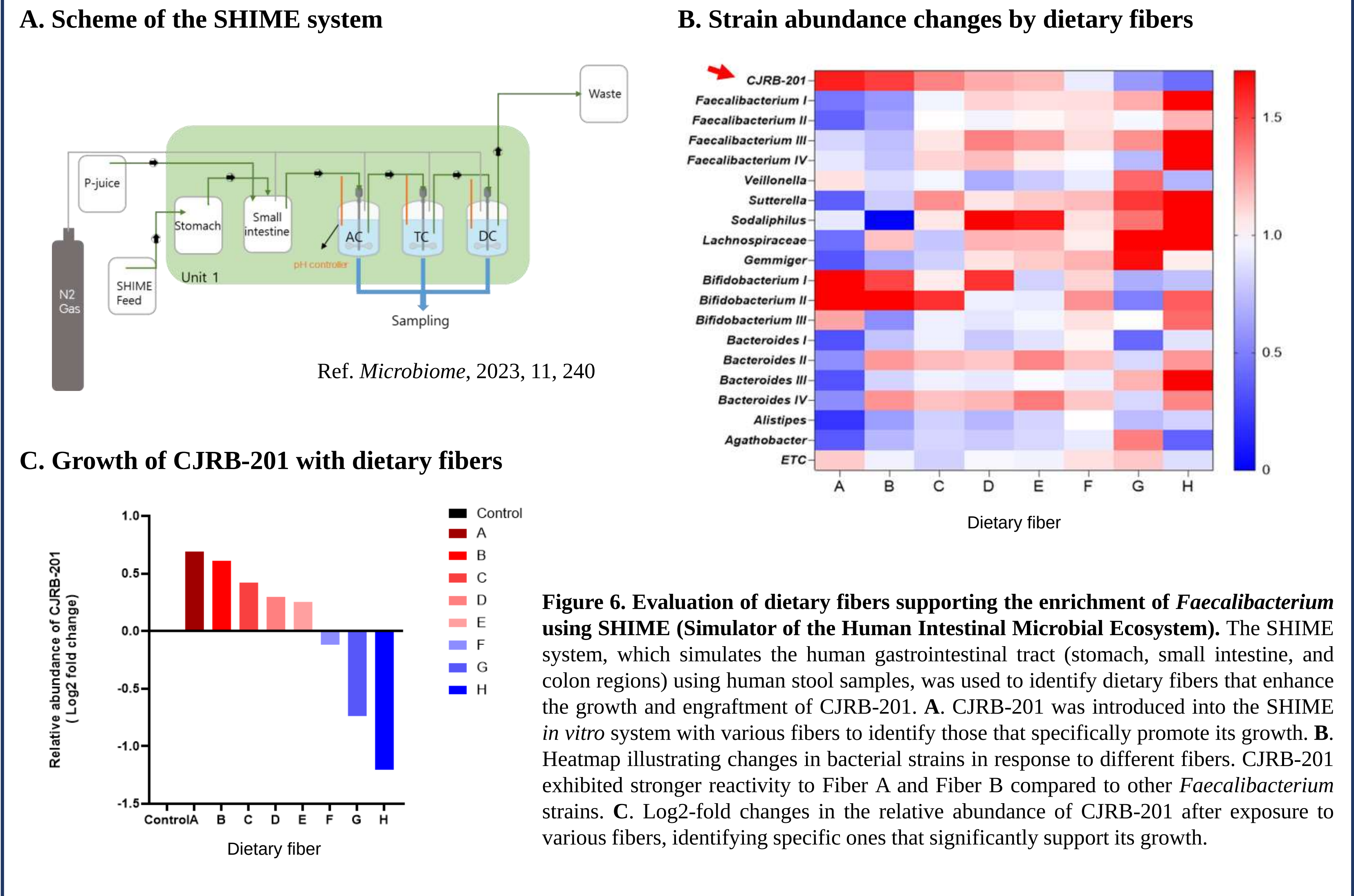
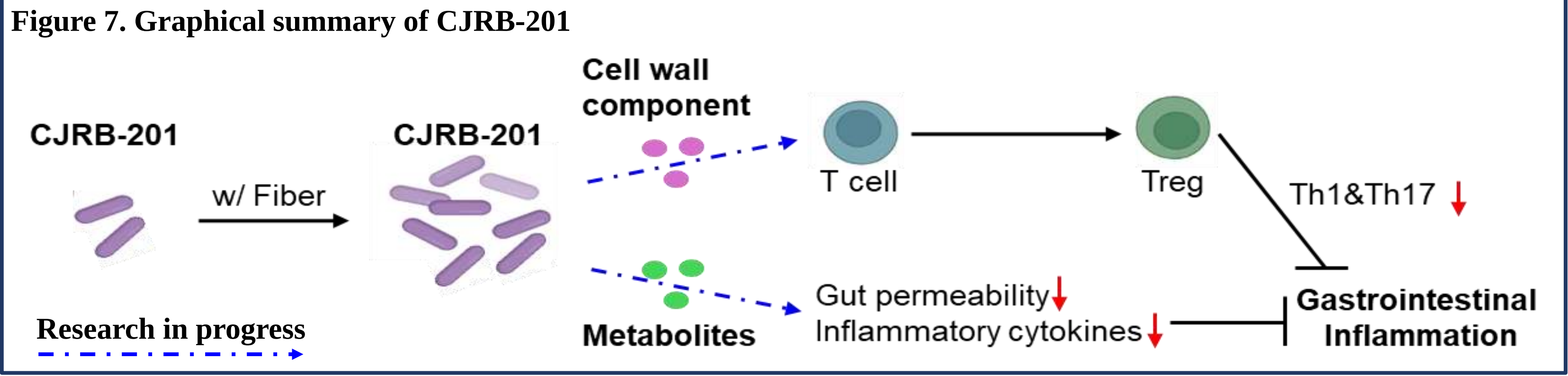


Figure 6. Evaluation of dietary fibers supporting the enrichment of *Faecalibacterium* using SHIME (Simulator of the Human Intestinal Microbial Ecosystem). The SHIME system, which simulates the human gastrointestinal tract (stomach, small intestine, and colon regions) using human stool samples, was used to identify dietary fibers that enhance the growth and engraftment of CJRB-201. **A.** CJRB-201 was introduced into the SHIME *in vitro* system with various fibers to identify those that specifically promote its growth. **B.** Heatmap illustrating changes in bacterial strains in response to different fibers. CJRB-201 exhibited stronger reactivity to Fiber A and Fiber B compared to other *Faecalibacterium* strains. **C.** Log2-fold changes in the relative abundance of CJRB-201 after exposure to various fibers, identifying specific ones that significantly support its growth.



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Conflict of Interest

The authors have no potential conflict of interest to disclose